

Vectors, Matrices, and Linear Maps

A Rigorous, Self-Contained University-Level Reference

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1 Foundational Prerequisites: Sets, Functions, and Relations

Before constructing linear algebra, we define the most primitive objects.

1.1 Sets

Definition 1.1 (Set). A set is a well-defined collection of distinct objects. The objects are called elements or members.

Example 1.1. $\mathbb{N} = \{1, 2, 3, \dots\}$ is the set of natural numbers. $\mathbb{R}$ is the set of all real numbers. Notation: $x \in S$ means “ x is an element of S ”. For instance, $2 \in \mathbb{N}$ but $\frac{1}{2} \notin \mathbb{N}$.

1.2 Functions (Maps)

Definition 1.2 (Function). A function (or map) $f : X \rightarrow Y$ assigns to *every* element $x \in X$ *exactly one* element $f(x) \in Y$.

- X is the domain.
- Y is the codomain.
- The image is $\text{Im}(f) = \{f(x) \mid x \in X\}$, a subset of Y .

Example 1.2. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = x^2$. The domain is $\mathbb{R}$, the codomain is $\mathbb{R}$, but the image is $[0, \infty)$ because squares are never negative.

2 Scalars and the Field

2.1 Field

Definition 2.1 (Field). A field $\mathbb{F}$ is a set equipped with two binary operations, addition (+) and multiplication ($\cdot$), satisfying the usual arithmetic rules: associativity, commutativity, distributivity, existence of additive and multiplicative identities (0 and 1), and existence of inverses for every non-zero element. **Convention:** In this text, we take $\mathbb{F} = \mathbb{R}$, the field of real numbers.

2.2 Scalar

Definition 2.2 (Scalar). A scalar is an element of the field $\mathbb{F}$. That is, a single real number.

Example 2.1. Let $\alpha = 3$ and $\beta = -0.5$. Both are scalars in $\mathbb{R}$. Their sum is $2.5 \in \mathbb{R}$, and their product is $-1.5 \in \mathbb{R}$. The field is closed under these operations.

3 The Vector — The Fundamental Element

Definition 3.1 (Coordinate Vector). For $n \in \mathbb{N}$, define $\mathbb{R}^n$ as the set of all ordered n -tuples of real numbers:

$$\mathbb{R}^n = \{(v_1, v_2, \dots, v_n) \mid v_i \in \mathbb{R}\}.$$

Each element $v = (v_1, \dots, v_n)$ is called a coordinate vector.

Example 3.1. Let $v = (3, -1, 2) \in \mathbb{R}^3$ and $u = (1, 4, -2) \in \mathbb{R}^3$.

- **Addition:** $v + u = (3 + 1, -1 + 4, 2 + (-2)) = (4, 3, 0)$.
- **Scalar Multiplication:** Let $\alpha = 2$. Then $\alpha v = (2 \cdot 3, 2 \cdot (-1), 2 \cdot 2) = (6, -2, 4)$.

Both results remain in $\mathbb{R}^3$. This property is called closure.

4 Vector Space — The Algebraic Structure

Definition 4.1 (Vector Space). A vector space over a field $\mathbb{F}$ is a set V (whose elements are called vectors) together with two operations:

- (i) **Addition:** $+: V \times V \rightarrow V$
- (ii) **Scalar Multiplication:** $\cdot: \mathbb{F} \times V \rightarrow V$

These operations must satisfy the following eight axioms for all $u, v, w \in V$ and all $\alpha, \beta \in \mathbb{F}$:

- (V1) $(u + v) + w = u + (v + w)$ (Associativity of addition)
- (V2) $u + v = v + u$ (Commutativity of addition)
- (V3) $\exists \mathbf{0} \in V : v + \mathbf{0} = v$ (Existence of additive identity)
- (V4) $\forall v, \exists (-v) \in V : v + (-v) = \mathbf{0}$ (Existence of additive inverses)
- (V5) $\alpha(u + v) = \alpha u + \alpha v$ (Distributivity over vector addition)
- (V6) $(\alpha + \beta)v = \alpha v + \beta v$ (Distributivity over scalar addition)

(V7) $\alpha(\beta v) = (\alpha\beta)v$ (Compatibility of scalar multiplication)

(V8) $1 \cdot v = v$ (Identity element of scalar multiplication)

Example 4.1. $\mathbb{R}^2$ with component-wise operations is a vector space. The zero vector is $(0, 0)$, inverses are $(-x, -y)$, and all other axioms follow from real-number properties.

5 Linear Independence, Span, and Basis (Fully Grounded)

5.1 Linear Combination and Span

Definition 5.1 (Linear Combination). A linear combination of vectors $v_1, v_2, \dots, v_k \in V$ is an expression of the form

$$\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_k v_k,$$

where $\alpha_i \in \mathbb{F}$.

Definition 5.2 (Span). The span of a set $S \subseteq V$, denoted $\text{span}(S)$, is the set of all finite linear combinations of vectors from S :

$$\text{span}(S) = \left\{ \sum_{i=1}^k \alpha_i v_i \mid v_i \in S, \alpha_i \in \mathbb{F}, k \in \mathbb{N} \right\}.$$

The span is the smallest subspace of V that contains S .

Example 5.1. In $\mathbb{R}^2$, let $S = \{(1, 0)\}$. Then $\text{span}(S) = \{(t, 0) \mid t \in \mathbb{R}\}$, which is exactly the x -axis.

5.2 Linear Independence

Definition 5.3 (Linear Independence). A set of vectors $\{v_1, \dots, v_k\} \subset V$ is linearly independent if the only solution to the homogeneous equation

$$\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_k v_k = \mathbf{0}$$

is the trivial solution $\alpha_1 = \alpha_2 = \dots = \alpha_k = 0$. If a non-trivial solution exists, the set is linearly dependent (meaning at least one vector is a linear combination of the others).

5.3 Basis — The Coordinate System of a Vector Space

Definition 5.4 (Basis and Its Fundamental Property). A basis for a vector space V is a set $\mathcal{B} \subset V$ that satisfies two conditions:

1. $\mathcal{B}$ is linearly independent.

2. $\text{span}(\mathcal{B}) = V$.

The most important consequence: Because a basis is linearly independent and spans the space, **every vector $v \in V$ can be written as a linear combination of the basis vectors in exactly one way.** This unique representation is what gives us a coordinate system. If the basis is $\mathcal{B} = \{b_1, b_2, \dots, b_n\}$, then the unique coefficients $(\alpha_1, \dots, \alpha_n)$ such that $v = \alpha_1 b_1 + \dots + \alpha_n b_n$ are called the coordinates of v with respect to $\mathcal{B}$.

Example 5.2 (Testing for a Basis). In $\mathbb{R}^3$, consider the set $\mathcal{B} = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$. This is the standard basis.

- It spans $\mathbb{R}^3$ because any vector (x, y, z) can be written as $x(1, 0, 0) + y(0, 1, 0) + z(0, 0, 1)$.
- It is independent because $x(1, 0, 0) + y(0, 1, 0) + z(0, 0, 1) = (0, 0, 0)$ forces $x = y = z = 0$.

Now consider $v_1 = (1, 0, 0)$, $v_2 = (0, 1, 0)$, $v_3 = (1, 1, 0)$. They are **not** a basis because they are dependent: $v_1 + v_2 - v_3 = 0$. They also do not span $\mathbb{R}^3$; any combination of them has zero as the third coordinate, so they only span the xy -plane.

6 Linear Maps — Structure-Preserving Functions

Definition 6.1 (Linear Map). Let V and W be vector spaces over the same field $\mathbb{F}$. A function $T : V \rightarrow W$ is called a linear map (or linear transformation) if for all $u, v \in V$ and all $\alpha, \beta \in \mathbb{F}$:

$$T(\alpha u + \beta v) = \alpha T(u) + \beta T(v).$$

In words, a linear map preserves addition and scalar multiplication.

6.1 Kernel and Image

Every linear map gives rise to two fundamental subspaces.

Definition 6.2 (Kernel and Image). Let $T : V \rightarrow W$ be a linear map.

- The kernel (or null space) of T is:

$$\ker(T) = \{v \in V \mid T(v) = \mathbf{0}_W\}.$$

The kernel measures the amount of information that is lost by the map.

- The image (or range) of T is:

$$\text{Im}(T) = \{T(v) \in W \mid v \in V\}.$$

The image measures the "reach" of the map.

Example 6.1. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be defined by $T(x, y) = (x + y, x - y)$.

- It is linear (proof omitted for brevity, but standard).
- $\ker(T) = \{(0, 0)\}$ because $x + y = 0$ and $x - y = 0$ imply $x = y = 0$.
- $\text{Im}(T) = \mathbb{R}^2$ because $T(1, 0) = (1, 1)$ and $T(0, 1) = (1, -1)$ are independent and span the whole space.

7 Matrix Representation

Definition 7.1 (Matrix of a Linear Map). Let $T : V \rightarrow W$ be a linear map. Let $\mathcal{E} = \{e_1, \dots, e_n\}$ be a basis for V , and $\mathcal{F} = \{f_1, \dots, f_m\}$ for W . For each e_j , we have $T(e_j) \in W$, so it can be uniquely written as:

$$T(e_j) = a_{1j}f_1 + a_{2j}f_2 + \dots + a_{mj}f_m.$$

The $m \times n$ matrix $A = [a_{ij}]$ is the matrix of T relative to the bases $\mathcal{E}$ and $\mathcal{F}$.

Example 7.1. Let $T(x, y) = (x + y, 2y)$. Its matrix relative to the standard basis is:

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}.$$

8 Rank, Nullity, and the Rank-Nullity Theorem

8.1 Definitions

Definition 8.1 (Rank and Nullity). Let $T : V \rightarrow W$ be a linear map.

- **Rank:** $\text{rank}(T) = \dim(\text{Im}(T))$. It is the number of independent directions in the output space that the map can actually produce.
- **Nullity:** $\text{null}(T) = \dim(\ker(T))$. It is the number of independent directions in the input space that get completely crushed to zero.

8.2 The Rank-Nullity Theorem

Theorem 8.1 (Rank-Nullity Theorem). Let $T : V \rightarrow W$ be a linear map. If V is finite-dimensional, then:

$$\dim(V) = \text{rank}(T) + \text{null}(T).$$

Equivalently, for an $m \times n$ matrix A : $n = \text{rank}(A) + \text{null}(A)$.

What Does This Theorem Really Say? (Intuition)

Imagine the domain V as a piece of clay. The linear map T acts like a press that flattens this clay. Some directions in the clay are completely squashed flat — they disappear into the zero vector. The dimension of this squashed part is the **nullity**. The remaining directions survive the pressing and form the final shape (the image). The number of independent directions in this final shape is the **rank**. The theorem simply states that the original number of independent directions (the dimension of V) is exactly the number of squashed directions plus the number of surviving directions. Nothing is lost in the counting; it's a perfect conservation law.

Detailed, Step-by-Step Proof (Fully Grounded)

Let $\dim(V) = n$. We will prove that the dimension of the domain equals the sum of the dimension of the image and the dimension of the kernel.

Step 1: Choosing a basis for the kernel (the squashed part). Let $k = \text{null}(T)$. By definition, the kernel $\ker(T)$ has dimension k . We choose a basis for this kernel:

$$\mathcal{U} = \{u_1, u_2, \dots, u_k\}.$$

These k vectors are linearly independent and they span exactly the set of all vectors that T sends to zero. *Why do we do this?* Because we want to isolate the part of the domain that completely disappears under the map.

Step 2: Extending to a basis for the whole space (adding the survivors). Since $\mathcal{U}$ is a linearly independent set in V , we can always add more vectors to it until it becomes a basis for the entire space V . Let us add exactly r vectors, named $w_1, w_2, \dots, w_r$, such that:

$$\mathcal{B} = \{u_1, \dots, u_k, w_1, \dots, w_r\}$$

is a basis for V . Because $\dim(V) = n$, we know that the total number of vectors in $\mathcal{B}$ must be n . Therefore:

$$k + r = n \implies r = n - k.$$

The vectors $w_1, \dots, w_r$ represent the part of the domain that is *not* in the kernel. These are the "survivors" that will map to form the image.

Step 3: Proving that the images of the survivors span the image. We claim that the set $\{T(w_1), T(w_2), \dots, T(w_r)\}$ covers the entire image of T . Take any output vector $y \in \text{Im}(T)$. By definition of the image, there must be some input $v \in V$ such that $T(v) = y$. Because $\mathcal{B}$ is a basis for V , we can write v uniquely as:

$$v = \alpha_1 u_1 + \dots + \alpha_k u_k + \beta_1 w_1 + \dots + \beta_r w_r.$$

Now apply the map T to both sides:

$$\begin{aligned} y = T(v) &= T\left(\sum_{i=1}^k \alpha_i u_i + \sum_{j=1}^r \beta_j w_j\right) \\ &= \sum_{i=1}^k \alpha_i T(u_i) + \sum_{j=1}^r \beta_j T(w_j). \end{aligned}$$

But recall that u_i are in the kernel, meaning $T(u_i) = 0$ for every single i . So the entire first sum vanishes entirely:

$$y = \sum_{j=1}^r \beta_j T(w_j).$$

What did we just show? We showed that every possible output vector y can be written as a combination of only the images of the survivor vectors. Hence these images span $\text{Im}(T)$.

Step 4: Proving that the images of the survivors are linearly independent.

We must check that there is no redundancy among the images of the survivors. Suppose there is a combination of them that equals zero:

$$\gamma_1 T(w_1) + \gamma_2 T(w_2) + \cdots + \gamma_r T(w_r) = 0.$$

Because T is linear, we can bring the scalars inside and combine everything into one application of T :

$$T(\gamma_1 w_1 + \gamma_2 w_2 + \cdots + \gamma_r w_r) = 0.$$

This equality means that the vector $\gamma_1 w_1 + \cdots + \gamma_r w_r$ belongs to the kernel of T . Since $\{u_1, \dots, u_k\}$ is a basis for the kernel, this vector must be expressible as a combination of the u_i 's. So there exist scalars $\delta_1, \dots, \delta_k$ such that:

$$\gamma_1 w_1 + \cdots + \gamma_r w_r = \delta_1 u_1 + \cdots + \delta_k u_k.$$

Bring all terms to the left side:

$$\gamma_1 w_1 + \cdots + \gamma_r w_r - \delta_1 u_1 - \cdots - \delta_k u_k = 0.$$

Now, here is the key observation: The set $\{u_1, \dots, u_k, w_1, \dots, w_r\}$ is a **basis**. A basis is linearly independent, meaning the *only* way to combine these vectors to get zero is if all coefficients are zero. Therefore, we must have:

$$\gamma_1 = \cdots = \gamma_r = 0, \quad \delta_1 = \cdots = \delta_k = 0.$$

Specifically, all the γ 's are zero. *What did we just prove?* We proved that $\{T(w_1), \dots, T(w_r)\}$ is linearly independent.

Step 5: Putting it all together to conclude. From Step 3, $\{T(w_1), \dots, T(w_r)\}$ spans the image. From Step 4, $\{T(w_1), \dots, T(w_r)\}$ is linearly independent. Therefore, these vectors form a **basis for the image**. This means the rank (the dimension of the image) is exactly the number of these vectors:

$$\text{rank}(T) = r.$$

From Step 2, we know that $k + r = n$. Since k is the nullity and n is the dimension of V , we can finally write:

$$\dim(V) = \text{null}(T) + \text{rank}(T).$$

The theorem is proven. □

A Quick Concrete Check: For the projection map $T(x, y) = (x, 0)$ on $\mathbb{R}^2$, we have $\text{rank} = 1$ (x-axis) and $\text{null} = 1$ (y-axis). The theorem gives $2 = 1 + 1$, which perfectly matches.

Example 8.1 (Applying the Theorem). *Let A be a 4×5 matrix with $\text{rank}(A) = 3$. Find its nullity and interpret it. Here the domain has dimension $n = 5$. By the Rank-Nullity Theorem, $\text{null}(A) = 5 - 3 = 2$. This means the homogeneous system $Ax = 0$ has a solution space of dimension 2 (a plane). The map is not one-to-one because it collapses an entire 2D plane to zero.*

9 Orthogonality (Fully Grounded)

Before discussing orthogonal projections, we must define what it means for vectors to be orthogonal.

9.1 Dot Product (Inner Product)

Definition 9.1 (Dot Product). *For vectors $u = (u_1, \dots, u_n)$ and $v = (v_1, \dots, v_n)$ in $\mathbb{R}^n$, their dot product (or inner product) is the scalar:*

$$u \cdot v = u_1v_1 + u_2v_2 + \dots + u_nv_n = \sum_{i=1}^n u_iv_i.$$

9.2 Orthogonality

Definition 9.2 (Orthogonal Vectors). *Two vectors u and v are called orthogonal (or perpendicular) if their dot product is zero:*

$$u \cdot v = 0.$$

Geometric meaning: The angle between the two vectors is exactly 90° . For example, in $\mathbb{R}^2$, the vectors $(1, 0)$ and $(0, 1)$ are orthogonal because $1 \cdot 0 + 0 \cdot 1 = 0$.

9.3 Orthogonal Projection

Definition 9.3 (Orthogonal Projection). *An orthogonal projection onto a subspace U is a linear map $P : V \rightarrow V$ such that:*

- $P(v) \in U$ for all $v \in V$ (the image is U).
- $v - P(v)$ is orthogonal to every vector in U (i.e., $v - P(v) \in U^\perp$).

In simpler terms: $P(v)$ is the "closest point" in U to v . The error vector $v - P(v)$ is perpendicular to the subspace U .

10 Canonical Example: Orthogonal Projection

Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear map that projects every vector orthogonally onto the x -axis:

$$T(x, y) = (x, 0).$$

Why is it orthogonal? The image is the x -axis. The kernel is the y -axis. Every vector in the y -axis is orthogonal to every vector in the x -axis (their dot product is zero). So $v - T(v) = (0, y)$ is indeed orthogonal to the x -axis.

Matrix Representation:

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}.$$

Kernel and Image:

$$\ker(T) = \{(0, y) \mid y \in \mathbb{R}\} \quad (\text{the } y\text{-axis}), \quad \dim = 1.$$

$$\text{Im}(T) = \{(x, 0) \mid x \in \mathbb{R}\} \quad (\text{the } x\text{-axis}), \quad \dim = 1.$$

Rank-Nullity Check:

$$\dim(\mathbb{R}^2) = 2, \quad \text{rank}(T) + \text{null}(T) = 1 + 1 = 2.$$

The theorem holds.

11 Change of Basis and Computing P (Fully Detailed)

When we change the coordinate system, we need a matrix that translates coordinates from the new basis to the old basis.

Definition 11.1 (Change-of-Basis Matrix P). *Let the "old" basis be $\mathcal{E} = \{e_1, \dots, e_n\}$, and the "new" basis be $\mathcal{E}' = \{e'_1, \dots, e'_n\}$. The change-of-basis matrix from $\mathcal{E}'$ to $\mathcal{E}$ is denoted by P . **How to construct P :***

1. Take the first new basis vector e'_1 .
2. Express e'_1 as a linear combination of the old basis vectors $\mathcal{E}$.
3. The coefficients of this expression become the entries of the **first column** of P .
4. Repeat this for e'_2 to get the second column, and so on.

In other words, the j -th column of P is the coordinate vector $[e'_j]_{\mathcal{E}}$.

Example 11.1 (Step-by-Step Calculation of P). *Let the old basis be the standard basis in $\mathbb{R}^2$: $\mathcal{E} = \{(1, 0), (0, 1)\}$. Let the new basis be $\mathcal{E}' = \{(1, 1), (1, -1)\}$. We want to find P .*

Step 1: Express the first new basis vector $(1, 1)$ in the old basis.

$$(1, 1) = 1 \cdot (1, 0) + 1 \cdot (0, 1).$$

So the coefficients are $(1, 1)$. This is the first column of P .

Step 2: Express the second new basis vector $(1, -1)$ in the old basis.

$$(1, -1) = 1 \cdot (1, 0) + (-1) \cdot (0, 1).$$

So the coefficients are $(1, -1)$. This is the second column of P .

Step 3: Assemble the matrix.

$$P = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

Now, P converts coordinates from the new basis to the old basis. If a vector has new coordinates (x', y') , its old coordinates are:

$$[x]_{\text{old}} = P \cdot [x]_{\text{new}}.$$

Step 4: Find the inverse to go the other way.

$$P^{-1} = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{bmatrix}.$$

If a vector has old coordinates $(3, 1)$, its new coordinates are:

$$[x]_{new} = P^{-1} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

Verification: $2(1, 1) + 1(1, -1) = (3, 1)$. ✓

12 Common Fallacies Resolved

Fallacy	Why It Is Wrong	The Formal Truth
$AB = BA$	Composition of functions is not commutative; order matters.	$AB \neq BA$ generally. AB means “apply B first, then A ”.
Column rank is different from row rank.	Common misconception.	Column rank and row rank are always equal .
Rank-Nullity applies in infinite dimensions.	Requires finite basis.	Only valid for $\dim(V) < \infty$.
Change of basis is always $P^{-1}AP$.	Only for endomorphisms ($V = W$).	General: $A' = Q^{-1}AP$.

13 Exercises with Solutions

Exercise 1

Let A be a 6×4 matrix with $\text{rank}(A) = 2$.

- (a) Find $\text{null}(A)$.
- (b) Can $Ax = b$ have a unique solution? Justify.

Solution: (a) $4 - 2 = 2$. (b) No, because nullity > 0 implies infinite solutions.

Exercise 2

Prove $\text{rank}(AB) \leq \min\{\text{rank}(A), \text{rank}(B)\}$. **Solution:** $\text{Im}(AB) \subseteq \text{Im}(A)$, so $\text{rank}(AB) \leq \text{rank}(A)$. Also, $\text{rank}(AB) = \dim(A(\text{Im}(B))) \leq \dim(\text{Im}(B)) = \text{rank}(B)$.

14 Why This Matters: Detailed Machine Learning Context

Linear algebra is the engine of ML.

- **Basis & Coordinates:** In ML, we constantly change bases to find better representations (e.g., PCA, word embeddings). A basis gives us a unique coordinate system for our data.
- **Rank:** The rank of a data matrix tells us how many independent features exist. Low rank means redundancy.
- **Nullity & Rank-Nullity:** In optimization, the nullity tells us how many flat directions exist in the loss landscape. This directly impacts the convergence of gradient descent.
- **Orthogonality:** Orthogonality is central to regularization (orthogonal features are independent) and to least-squares solutions, where the residual is orthogonal to the column space.
- **Change of Basis (P):** PCA is a change of basis to the eigenvectors. Word2Vec embeds discrete words into a continuous basis where semantic relationships become simple arithmetic.

Complete Glossary

Conclusion

This completes the self-contained, axiomatically rigorous exposition. Every term is now fully grounded: Basis has its unique representation property, P has a clear construction rule, Orthogonality is defined via the dot product, and the Rank-Nullity proof is broken down into intuitive, human-readable steps with a clear "why" for every action.

Term	Formal Definition	Key Intuition
Scalar	Element of $\mathbb{F}$ (e.g., $\mathbb{R}$).	Single number for scaling.
Vector	Element of a vector space V .	Data point, arrow, tuple.
Vector Space	Set with $+$ and $\cdot$ satisfying 8 axioms.	Container for data.
Linear Independence	Only trivial solution to $\sum \alpha_i v_i = 0$.	No redundancy.
Basis	Linearly independent spanning set.	Unique coordinate system.
Linear Map	$T(\alpha u + \beta v) = \alpha T(u) + \beta T(v)$.	Neural network layer.
Kernel	$\ker(T) = \{v \mid T(v) = 0\}$.	Information lost.
Image	$\text{Im}(T) = \{T(v) \mid v \in V\}$.	Reachable outputs.
Rank	$\dim(\text{Im}(T))$.	Number of independent features.
Nullity	$\dim(\ker(T))$.	Redundancy / free variables.
Rank-Nullity	$\dim(V) = \text{rank}(T) + \text{null}(T)$.	Conservation of dimensions.
Orthogonal	$u \cdot v = 0$.	Perpendicular directions.
Change of Basis	$A' = Q^{-1}AP$.	Reinterpreting data (PCA).